saprophytic organisms. Some angiosperms also follow saprophytic mode of nutrition. Example: *Neottia*. Roots of *Neottia* (Bird's Nest Orchid) associate with mycorrhizae and absorb nutrients as a saprophyte. *Monotropa* (Indian Pipe) grow on humus rich soil found in thick forests. It absorbs nutrient through mycorrhizal association (Figure 12.9).



Neottia (Bird's Nest Orchid) ***Monotropa*** (Indian Pipe)

Figure 12.9: Saprophytic Mode of nutrition

12.9.2 Parasitic mode of nutrition in angiosperms

Organisms deriving their nutrient from another organism (host) and causing disease to the host are called parasites.

a. Obligate or Total parasite - Completely depends on host for their survival and produces haustoria.

i. **Total stem parasite**: The leafless stem twine around the host and produce haustoria. Example: *Cuscuta* (Dodder), a rootless plant growing on *Zizyphus*, *Citrus* and so on.

ii. **Total root parasite**: They do not have stem axis and grow in the roots of host plants produce haustoria. Example: *Rafflesia*, *Orobanche* and *Balanophora*.

b. Partial parasite - Plants of this group contain chlorophyll and synthesize carbohydrates. Water and mineral requirements are dependent on host plant.

i. **Partial Stem Parasite**: Example: *Loranthus* and *Viscum* (Mistletoe)

Loranthus grows on fig and mango trees and absorb water and minerals from xylem.

ii. **Partial root parasite**: Example: *Santalum album* (Sandal wood tree) in its juvenile stage produces haustoria which grows on roots of many plants (Figure 12.10).

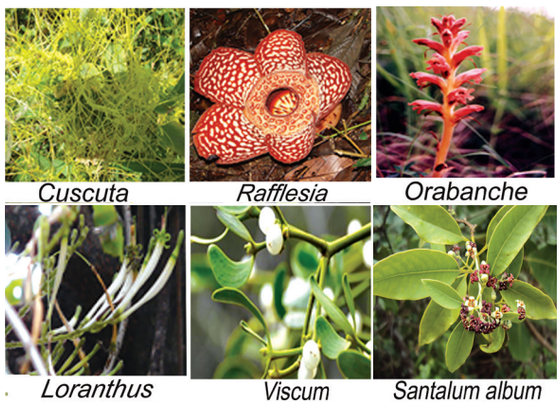


Figure 12.10: Parasitic Mode of Nutrition

12.9.3 Symbiotic mode of Nutrition

a. **Lichens**: It is a mutual association of Algae and Fungi. Algae prepares food and fungi absorbs water and provides thallus structure.

b. **Mycorrhizae**: Fungi associated with roots of higher plants including Gymnosperms. Example: *Pinus*.

c. ***Rhizobium* and Legumes**: This symbiotic association fixes atmospheric nitrogen

d. **Cyanobacteria and Coralloid Roots**: This association is found in *Cycas* where *Nostoc* associates with its coralloid roots. (Figure 12.11).

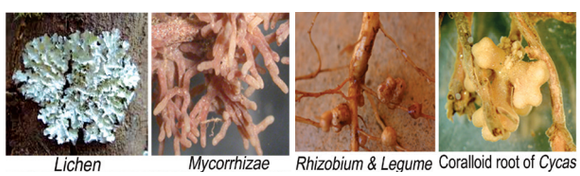


Figure 12.11: Symbiotic mode of nutrition

12.9.4 Insectivorous mode of nutrition

Plants which are growing in nitrogen deficient areas develop insectivorous habit to resolve nitrogen deficiency. These plants obtain nitrogen from the insects

a. ***Nepenthes*** (Pitcher plant): Pitcher is a modified leaf and contains digestive enzymes. Rim of the pitcher is provided with nectar glands and acts as an attractive

XIth_Botany_Chapter_12_EM.indd 268 5/17/2020 9:19:41 PM